

Tip of the Finger or Tip of the Tongue?

The Effects of Verbal Communication on Online Multi-Player Team Performance

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ABSTRACT

Virtual environments are increasingly being used to support team training. However, despite their growing ubiquity, less is known on how task difficulty in such environments mitigates verbal communication patterns. We report on a study investigating this research question, finding that although manipulations of task difficulty did appear to impact broad team level communication measures (e.g. how much they talked), it largely had minimal impact on team communication dynamics (e.g. the flow of communication between various roles in the teams). The implications of this study for the design of future team training environments along with the possibility of artificial agents to support team training as a novel use case are then discussed in turn.

CCS CONCEPTS

• **Human-centered computing** → Collaborative and social computing; Empirical studies in collaborative and social computing; • **Applied computing** → Education; Interactive learning environments.

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KEYWORDS

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1 INTRODUCTION

Virtual environments are increasingly being used to train human teams, particularly in scenarios where real-world training would otherwise be dangerous, expensive or where teams are geographically dispersed. Such environments have been successfully used to facilitate training in a variety of diverse contexts from search-and-rescue [1] to aerial reconnaissance [2]. These potential benefits of virtual training environments have been exemplified during the ongoing COVID-19 pandemic where face-to-face interaction is not feasible. For instance, the online video conferencing platform Zoom reported hosting approximately 300 million participants daily [9].

Despite recent advancements in connectivity, realism of extended reality environments, and adoption of collaborative technology, one largely unexplored question is the role verbal communication in such environments has on effective, coordinated team performance. On the one hand, in the absence of non-verbal communication found in physically collocated environments, one may predict that verbal communication would play an increasing role.

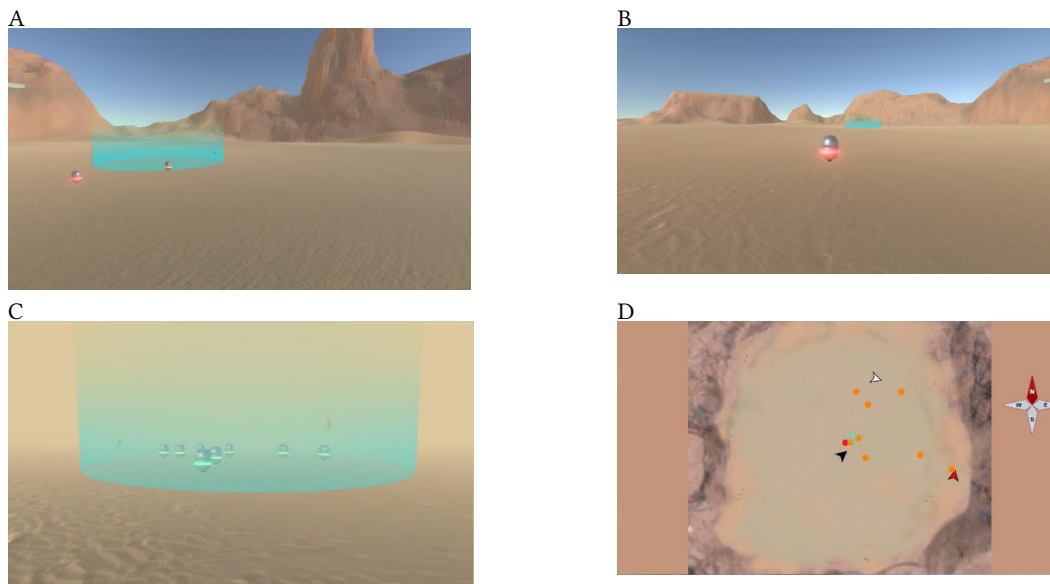


Figure 1: Desert Herding environment. (a) ground player's view in a clear condition; (b) ground player chasing an autonomous agent towards the containment area; (c) ground player's view in the foggy condition; (d) operator's map view of the environment

Indeed, the fields of psychology and cognitive science have shown that verbal communication plays a fundamental role in a variety of real-world team contexts from the safe landing of an airliner carrying hundreds of passengers [4] to winning competitive team sports [3, 6]. However, verbal communication in virtual environments may be unhelpful if participants struggle to build common ground to communicate effectively during training.

This work contributes to the burgeoning field of designing virtual environments to support effective team collaboration and learning. First, we report on a preliminary study which investigated the relationship between verbal communication and task difficulty within the context of a virtual multiplayer task environment. Second, we discuss the implications of the findings on the design of future virtual training environments. Lastly, we discuss several potential avenues for future work in this growing body of work.

2 CURRENT STUDY

To examine the role verbal communication has on team performance in the context of a virtual multiplayer environment, we designed a virtual environment called Desert Herding designed using Unity3D (further detail on this environment can be found in [5]). In this virtual environment, teams of four participants were tasked with locating and ultimately containing *autonomous agents* into a containment zone (Figure 1). Three of the participants were given the role of a *ground-player* who could navigate in the virtual world (Figures 1-1). The fourth participant was given the role of *operator* who had a birds-eye-view of the environment but could not directly interact with the autonomous agents (Figure 1). Participants had up to 5 minutes to successfully contain all *autonomous agents* within the containment area for a period of five consecutive

seconds. During the experiment session, participants had to complete 16 trials allowing them to experience all sets of experiment conditions twice: *visibility* (e.g., high visibility of 150m vs. low visibility of 10m), number of autonomous agents to contain (e.g., 9 vs 18) and *perturbation* (whether an agent appears towards the end of the trial). All participants had an audio channel to speak to each other during the trials. A total of 40 participants were recruited forming ten teams of four players. Participants received \$30 AUD for their time and had the opportunity to receive an additional \$50 AUD if their team had the best performance. The study was approved by the Macquarie University Institutional Review Board.

2.1 Results

Upon completion of the experimental session, the recorded audio from each team's completed trials were transcribed by the authors and research assistants. Two of the authors then coded the transcribed audio across two sets of codes using NVivo 12. One set of codes was used to determine whether the participant's utterances were related to the task (e.g., ON_TASK vs OFF_TASK). The second set of codes determine who the speaker in the transcript was directing their utterances towards (e.g., operator directing their speech at a specific ground player). Both sets of codes showed strong inter-rater reliability with a kappa score of 0.92 and 0.87 respectively. Aside from in-trial audio, team performance data was also collected, namely the duration of each trial (in seconds), *TDuration*.

Several measures for analyzing the transcripts and coded communication data were developed, which will be briefly discussed in turn. Firstly, a measure of the number of utterances (e.g. words) participants spoke during a trial, normalized by *TDuration* referred to as conversation density (*CDensity*) was computed to get a sense

Table 1: Mean (Standard Deviation) Team Measures by Condition

Measure	Perturbation				No Perturbation			
	Fog		No Fog		Fog		No Fog	
	9 Bots	18 Bots	9 Bots	18 Bots	9 Bots	18 Bots	9 Bots	18 Bots
TDuration	192.77 (37.37)	259.23 (33.81)	159.16 (23.51)	229.85 (18.66)	165.54 (21.57)	237.87 (29.68)	123.21 (13.60)	183.31 (26.04)
CDensity	2.26 (0.63)	2.35 (0.66)	1.96 (0.71)	1.91 (0.79)	2.31 (0.65)	2.33 (0.60)	1.80 (0.68)	1.76 (0.80)
OnTask	0.95 (0.09)	0.98 (0.03)	0.90 (0.15)	0.89 (0.15)	0.96 (0.06)	0.94 (0.13)	0.90 (0.15)	0.89 (0.25)
ETP	0.22 (0.07)	0.15 (0.05)	0.17 (0.07)	0.21 (0.09)	0.18 (0.05)	0.18 (0.05)	0.24 (0.13)	0.19 (0.10)
TTO	0.75 (0.14)	0.76 (0.12)	0.67 (0.20)	0.67 (0.21)	0.75 (0.14)	0.77 (0.16)	0.75 (0.16)	0.70 (0.23)
TTGP	0.73 (0.21)	0.79 (0.15)	0.57 (0.26)	0.53 (0.15)	0.69 (0.13)	0.76 (0.17)	0.68 (0.21)	0.61 (0.25)

of how “busy,” the team communication channel was during the trial. Second, the proportion of total utterances which were coded as being ON_TASK, was computed (*OnTask*), to determine how focused team’s communication was on the task at hand. Thirdly, we computed how evenly distributed a participant’s utterances were to other participants, equality in talking partners (*ETP*), using the following formula adapted from [8]:

$$ETP = \frac{|\tau(P_i, P_j) - \tau(P_i, P_k)|}{\tau(P_i, P_j) + \tau(P_i, P_k)} + \frac{|\tau(P_i, P_k) - \tau(P_i, P_m)|}{\tau(P_i, P_k) + \tau(P_i, P_m)} + \frac{|\tau(P_i, P_j) - \tau(P_i, P_m)|}{\tau(P_i, P_j) + \tau(P_i, P_m)}$$

where the term $\tau(P_i, P_j)$ represents the proportion of the transcript where participant i’s speech was directed towards participant j. The fourth communication measure, total talking by operator (*TTO*), represents the proportion of total utterances where the operator was speaking. Lastly, total talking by ground players (*TTGP*) represents the proportion of ground player utterances that were directed towards the operator. Each team performance and communication measure were analyzed using 2 (Target Number: 9 vs. 18) $\times$ 2 (Visibility: no-Fog vs. Fog) $\times$ 2 (Target Perturbation: 0 vs 1) repeated measures ANOVAs, with Bonferroni corrections employed for any post-hoc analysis. Where assumptions of sphericity were violated, Greenhouse-Geisser corrections were employed. For some measures (e.g., trial duration), the assumption of normality was also violated. However, these violations were minimal (the result of ceiling or floor effects) and, thus, given the robustness of ANOVA to violations of normality the data was not transformed prior to analysis. A summary of the results can be found in Table 1.

As expected, all three task manipulations significantly influenced team performance. More specifically, the analysis of *TDuration* revealed significant main effects of *Target Number*, $F(1,9) = 308.32$, $p < .001$, $\eta_p^2 = .97$, *Visibility* $F(1,9) = 43.95$, $p < .001$, $\eta_p^2 = .83$, and *Target Perturbation*, $F(1,9) = 91.06$, $p < .001$, $\eta_p^2 = .91$, but no significant two-way interactions (all $F(1,9) < 4.56$, $p > .061$, $\eta_p^2 < .34$), nor three-way interactions, $F(1,7) = .829$, $p = .360$, $\eta_p^2 = .094$.

In terms of team communication measures, several statistically significant findings were discovered. First, an analysis of *CDensity*

as a function of conditions resulted in a significant main effect of *Visibility*, $F(1,9) = 59.62$, $p < .001$, $\eta_p^2 = .87$, and *Visibility* $\times$ *Target Perturbation* interaction, $F(1,9) = 6.699$, $p = .029$, $\eta_p^2 = .43$. Post-hoc analysis, however, revealed there was a significant difference in *CDensity* between the low ($M = 2.30$, $SD = 0.64$) and high ($M = 1.94$, $SD = 0.74$) visibility conditions ($p < .01$) when target perturbations were induced, as well as between low ($M = 2.32$, $SD = 0.62$) and high ($M = 1.79$, $SD = 0.71$) visibility conditions ($p < .001$) for non-perturbation trials, indicating that teams communicated significantly more during low visibility (fog trials), irrespective of perturbation (and target number). However, no significant main effects were observed for *Target Number*, *Target Perturbation*, nor for any interaction effects (all $F(1,9) < 1.53$, $p > .08$, $\eta_p^2 < .20$). Second, an analysis of on-task communication revealed a marginal effect for *Visibility*, $F(1,9) = 4.93$, $p = .054$, $\eta_p^2 = .35$, with more on-task information tending to be observed during low visibility (Fog) conditions compared to high visibilities (no-Fog) conditions. However, there were no other significant main effects nor interaction effects (all $F(1,9) < 1.35$, $p > .274$, $\eta_p^2 < .14$). Third, an analysis of *TTO* revealed no main effects (all $F(1,9) < 4.24$, $p > .058$, $\eta_p^2 < .35$) nor two or three-way interaction (all $F(1,9) < 3.25$, $p > .104$, $\eta_p^2 < .27$), indicating that the operator exhibited the same relative magnitude (proportion) of communication across all conditions. Fourth, when investigating how evenly the operator communicated between the three ground players, the analysis of *ETP* also revealed no main effects for *Visibility* $F(1,9) = .58$, $p = .465$, $\eta_p^2 = .06$, *Target Number* $F(1,9) = 4.38$, $p = .066$, $\eta_p^2 = .33$, and *Target Perturbation*, $F(1,9) = .53$, $p = .486$, $\eta_p^2 = .06$, nor were there any interaction effects (all $F(1,9) < 2.92$, $p > .121$, $\eta_p^2 < .25$), indicating that on average operators tended to verbally engage with each ground player in their team equally.

Finally, the analysis of *TTGP* revealed a significant main effect of *Visibility*, $F(1,9) = 50.55$, $p < .001$, $\eta_p^2 = .85$, as well as a significant *Target Number* $\times$ *Visibility* interaction, $F(1,9) = 5.35$, $p = .046$, $\eta_p^2 = .373$. Post-hoc analysis resulted in a significant difference in the proportion of ground communication between low and high visibility for the 18-target condition ($p < .001$), but no significant difference between low and high visibility for the 9-target condition, indicating

that while overall a greater proportion of ground players communication was directed towards the operator during low visibility trials compared to high visibility trials, this was most pronounced the more targets that needed to be herded. No other main nor interaction effects (all $F(1,9) < 3.95$, $p > .078$, $\eta^2 < .305$) were found.

3 DISCUSSION

The present study has demonstrated the role verbal communication plays in mediating team performance and coordination in virtual training environments. In particular, increases in task difficulty led to greater use of verbal communication (e.g., *CDensity*). A potential implication for designers is to allow participants to engage in verbal communication which is a capability often absent in many virtual training environments. In addition, not all training interventions impacted communication measures equally. For instance, although a reduction in visibility increased the amount of communication it did not have a significant effect on how evenly the operator spoke to ground players (e.g. *TTO*). Therefore, careful selection of training interventions is crucial if virtual training environments are to be used to encourage desired types of communication behaviors.

Looking ahead, there are several avenues for future research. In particular, performing the same study in the absence of the operator would enable us to determine whether role dynamics (e.g. operator vs ground-players) impacted the rate and flow of verbal communication amongst team members. Relatedly, performing the same study in the absence of verbal communication would allow us to directly measure how verbal communication contributes to team performance.

Lastly, an ongoing interest of the lab is the development of Artificial Agents (AAs) whose potential benefits of are manifold. First, AAs would be able to fill in for participants unable to attend, fulfilling the promise of virtual training environments to allow geographically dispersed teams to train anytime and anywhere. Second,

AAs would enable teams to be formed with the right mixture of expertise. Third, AAs could provide team members with frank and honest feedback during training. Indeed, prior work has discussed some of these benefits in detail[7].

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